System Analysis

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Project Managment

The science (or art) of project management is in making _____ of size, time and cost.

- Benchmark comparisons
- Analytical and educated estimates
- Trade-offs
- Maximum calculations
- Minimum calculations

The most common reason for schedule and cost overruns is _____

- Team conflict
- Lack of communication from project manager to project team
- Lack of support by sponsor and champion
- Scope creep
- Adding people to a late project

Micah is a fairly new project manager. He estimated for a project plan (on the planning phase) that the project would cost $50,000 and take 20 weeks. According to the margin of error guidelines for well-done estimates, that could range from:

- $0 and $100,000 – and between 15 and 25 weeks
- $10,000 and $60,000 – and between 12 and 28 weeks
- $0 and $100,000 – and between 0 and 40 weeks
- $5,000 and $100,000 – and between 10 and 30 weeks
- $25,000 and $75,000 – and between 10 and 30 weeks

Mya is managing a project. Unfortunately, the entire project team lost four days due to a hurricane in the area and a couple of the project team lost even more time when dealing with insurance claims and rebuilding. At the six-week mark of a nine-week project, she figures the team is one week behind. What might be the BEST solution?

- Add several people to the project to get it back to schedule
- Do not change the schedule, but expect the project and development teams to work longer days and every-other weekend.
- Since the project is about 16% behind, increase the project to about 10-and-one-half weeks
- Since the project is about 20% behind, increase the project to about 12 week.
- Add three people to the project – AND – increase the overall project by one week (to a total of ten weeks)

STAFFING THE PROJECT

Staffing the project includes:

- Determining how many people should be assigned to the project
- Matching people's skills with the needs of the project
- Motivating them to meet the project's objectives
- Minimizing the conflict that will occur over time

The deliverables for this part of project management **are a staffing plan**, which describes the number and kinds of people who will work on the project, the **overall reporting structure**, and the **project charter**, which describes the project's objectives and rules.

Jelled team

According to DeMarco and Lister*, “[a] jelled team is a group of people so strongly knit that the whole is greater than the sum of the parts. The production of such a team is greater than that of the same people working in unjelled form.” They go on to state that a jelled “team can become almost unstoppable, a juggernaut for success

Demarco and Lister identify five characteristics of a jelled team.

* T. DeMarco and T. Lister, Peopleware: Productive Projects and Teams, 2nd Ed., p. 123.

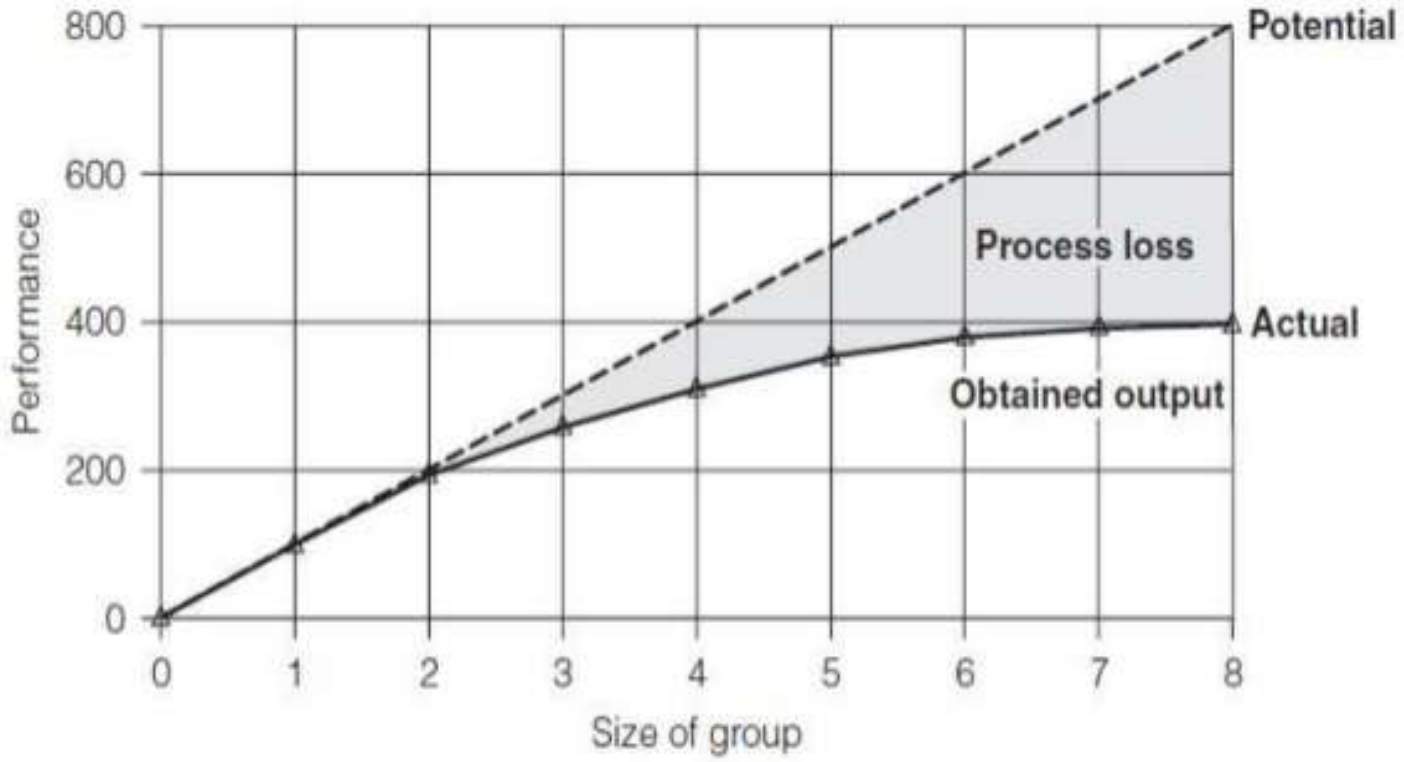
Ringelmann effect

The Ringelmann effect is the tendency for individual members of a group to become increasingly less productive as the size of their group increases. This effect, discovered by French agricultural engineer Maximilien Ringelmann (1861–1931), illustrates the inverse relationship that exists between the size of a group and the magnitude of group members' individual contribution to the completion of a task.



A tug of war is often used as an illustration of the Ringelmann effect. As more people are involved in a task, their average performance decreases, each participant tending to feel that their own effort is not critical.

Ringelmann Effect



Characteristics of a Jelled Team.

- **First**, jelled teams have a very low turnover during a project.

Typically, members of a jelled team feel a responsibility to the other team members. This responsibility is felt so intensely that for a member to leave the team, the member would feel that they were letting the team down and that they were breaking a bond of trust.

- **Second**, jelled teams have a strong sense of identity.

In many classes, when you are part of a group, the group chooses some cute name to identify the group and differentiate it from the other groups.

- **Third**, the strong sense of identity tends to lead the team into feeling a sense of eliteness.

The members of a jelled development team almost have a swagger about the way they relate to nonteam employees.

- **Fourth**, during the development process, jelled teams feel that the team owns the information system being developed and not any one individual member.

In many ways, you could almost say that jelled teams are a little communistic in nature. By this we mean that the individual contributions to the effort are not important to a true team. The only things that matter are the output of the team. However, this is not to imply that a member who does not deliver their fair share will not go unpunished. In a jelled team, any member who is not producing is actually breaking their bond of trust with the other team members.

- **The final** characteristic of a jelled team is that they really enjoy (have fun) doing their work.

The members actually like to go to work and be with their team members. Much of this can be attributed to the level of challenge they receive. If the project is challenging and the members of the team are going to learn something from completing the project, the members of a jelled team will enjoy tackling the project.

Ten Steps To More Productive Development Teams

1. Give Them Their Own Space

It is virtually impossible for a team to truely jell if its members are not all in one place. A lot of companies put their workers in individual cubicles that seem deliberately designed to stop them working effectively - they are shielded from their colleagues well enough to stop them communicating easily, but not well enough to stop them being distracted.

2. Try Agile Development Methods

3. Maximise Communication, Minimise Bureaucracy

A key to creating a high performance team is open, honest, face-to-face communication. Here are some ways to improve communication:

Have daily standup meetings so that each team member knows what the others are doing that day.

Encourage Pair Programming - two developers working together on a single PC.

Have group design sessions round a whiteboard instead of people working in isolation with a CASE tool.

4. No More Heroes

Discourage programmer heroics. Many teams have a 'lone ranger' who works in secrecy churning out large volumes of complex code that no-one else understands.

5. Let Them Grow

6. Give them Play Time and Rest Time

7. Trust the Team

8. Work Iteratively and Incrementally

9. Use Test First Design

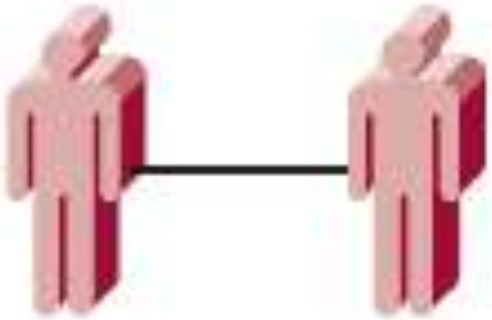
Test First Design (sometimes called Test Driven Development) means that you write the tests before you write the code to make the test pass.

10. Get A Team Coach

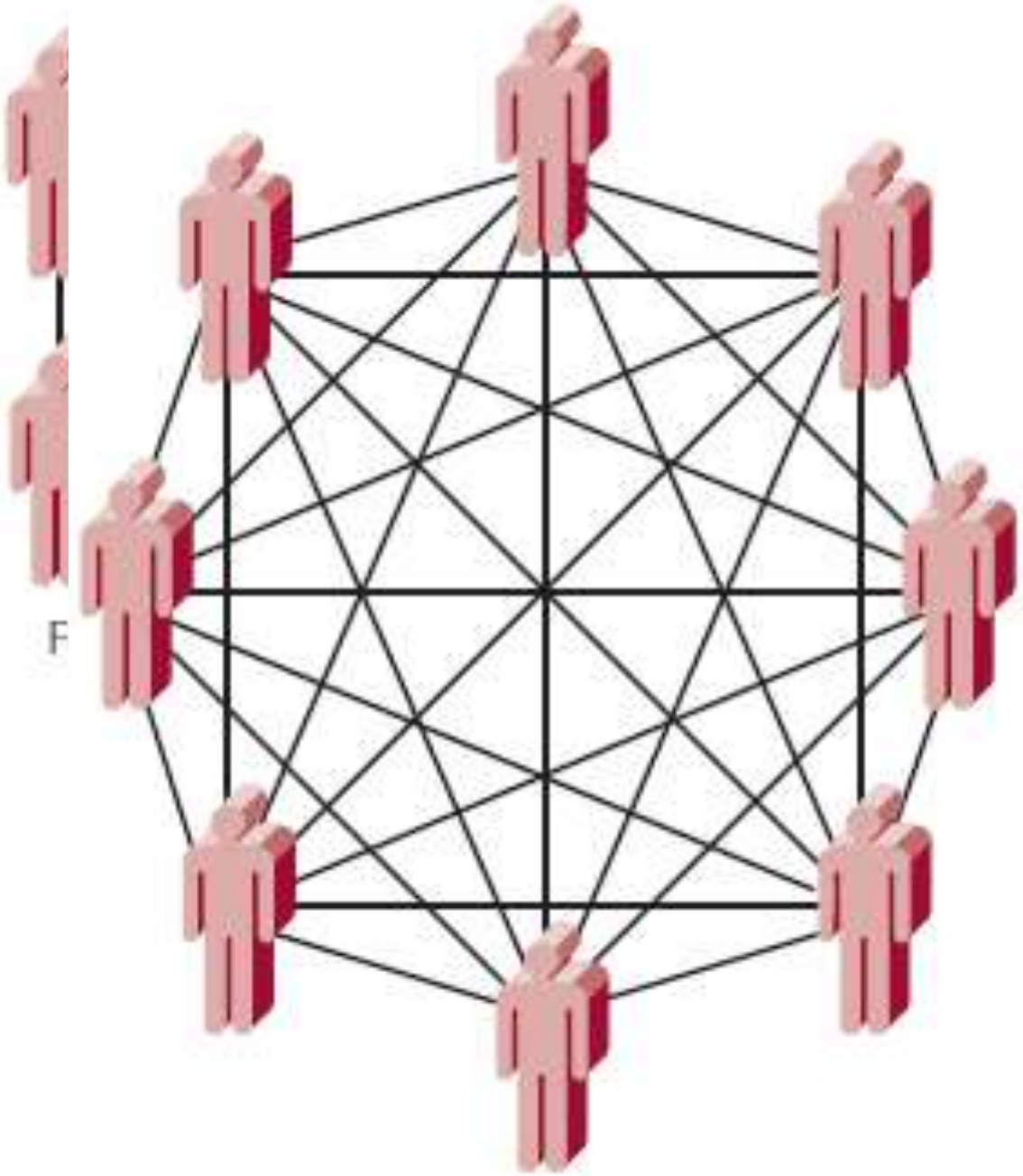
Staffing Plan

The **first step** to staffing is **determining the average number of staff needed for the project**. To calculate this figure, divide the total person-months of effort by the optimal schedule. So to complete a forty-person-month project in ten months, a team should have an average of four full-time staff members, although this may change over time as different specialists enter and leave the team (e.g., business analysts, programmers, technical writers).

Adding staff resources does not translate into increased productivity; staff size and productivity share a disproportionate relationship, mainly because it is more difficult to coordinate a large number of staff members. The more a team grows, the more difficult it becomes to manage.



Two-person team



Eight-person team

One way to reduce efficiency losses on teams is to understand the complexity that is created in numbers and to build in a reporting structure that tempers its effects. The general rule is to keep team sizes to fewer than eight to ten people; therefore,

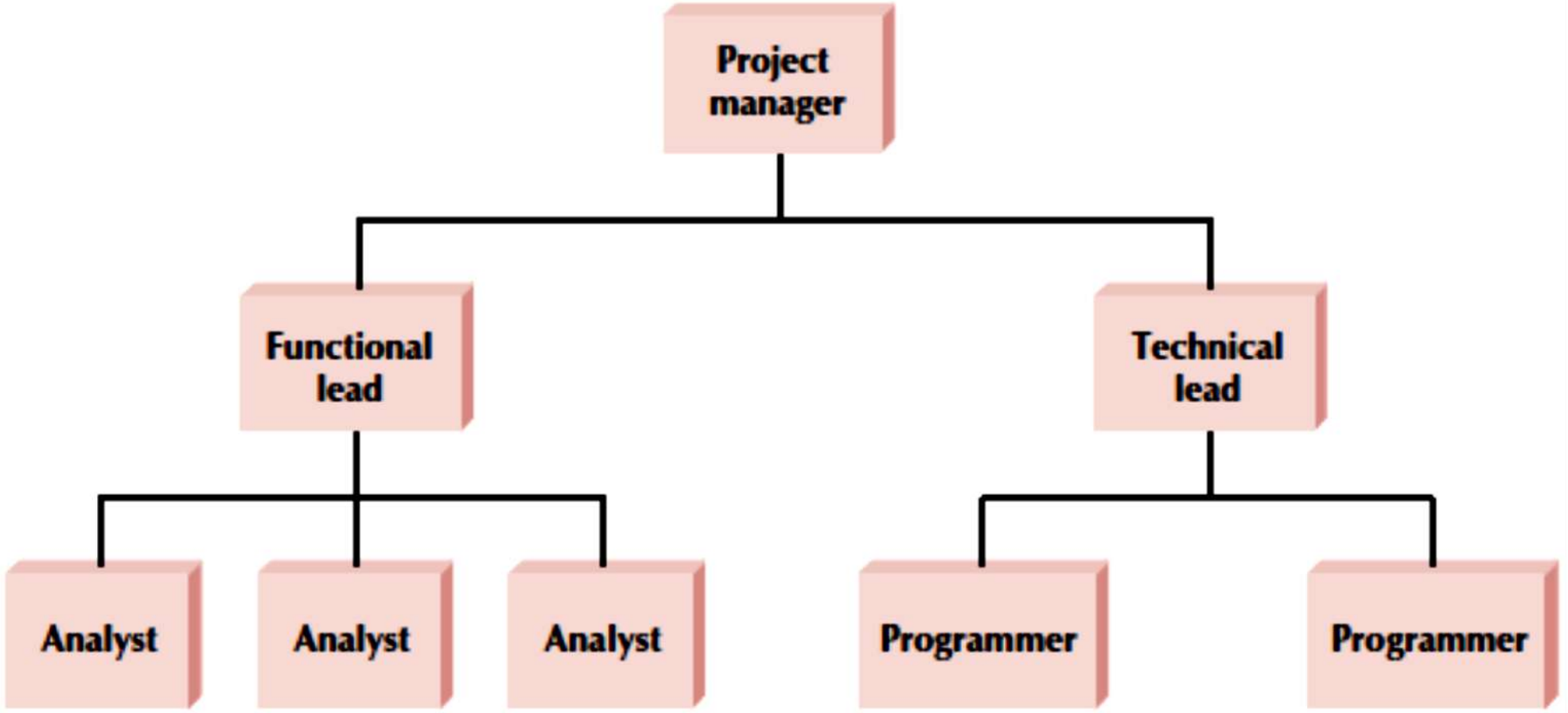
if more people are needed, create sub-teams. In this way, the project manager can keep the communication effective within small teams, which, in turn, communicate to a contact at a higher level in the project

Staffing Plan

After the project manager understands how many people are needed for the project, he or she creates a ***staffing plan*** that lists the roles and the proposed reporting structure that are required for the project. Typically, a project has one project manager, who oversees the overall progress of the development effort, with the core of the team comprising the various types of analysts. A ***functional lead*** is usually assigned to manage a group of analysts, and a **technical lead** oversees the progress of a group of programmers and more technical staff members.

There are many structures for project teams; Next Figure illustrates one possible configuration of a project team.

Possible configuration of a project team.



When you make assignments, remember that people have *technical skills* and *interpersonal skills*, and both are important on a project.

Motivation

Assigning people to tasks isn't enough; project managers need to motivate the people to make the project a success. *Motivation* has been found to be the number one influence on people's performance, but determining how to motivate the team can be quite difficult.

Types of Motivation

There are two main types of motivation – extrinsic and intrinsic.

Extrinsic motivation is when you use external factors to encourage your team to do what you want. Pay raises, time off, bonus checks, and the threat of job loss are all extrinsic motivators – some positive, some less so.

Intrinsic motivation is internal. It's about having a personal desire to overcome a challenge, to produce high-quality work, or to interact with team members you like and trust. Intrinsically motivated people get a great deal of satisfaction and enjoyment from what they do.

Types of Motivation

1

Intrinsic Motivation

- Attitude Motivation
- Competence Motivation
- Achievement Motivation
- Physiological Motivation
- Creative Motivation
- Introjected Motivation

2

Extrinsic Motivation

- Fear Motivation
- Incentive Motivation
- Social Motivation
- Power Motivation
- Identified Motivation

https://habr.com/ru/companies/psb/articles/938116/

You might think that good project managers motivate their staff by rewarding them with money and bonuses, but most project managers agree that this is the last thing that should be done. **The more often managers reward team members with money, the more they expect it—and most times monetary motivation won't work.** There are a set of principles to follow to motivate individuals in twenty-first century firms.

First rule suggests that 20 percent of an employee's time should be spent on some idea in which they believe. The project does not have to be related in any way to the project at hand. On the surface, this sounds like a colossal waste of time, but you should not throw this idea away. Have you used Gmail? This is how Google's Gmail and Google News became software products. If 20 percent sounds too high, Some authors suggests that you consider 10 percent to begin with.

Another recommendation that firms should be willing to **fund small "Now That" awards**. These awards are given as small signs of appreciation for doing a great job. However, these awards are not given by a manager to an employee but from an employee to a peer of the employee. The awards are monetary, but they are very small, typically $50. As such, they really are not relevant from a monetary perspective. However, they are very relevant because they are given by one of the employee's colleagues to show that some action that the employee did was appreciated.

Next, management should periodically consider giving each employee a day on which they can work on anything they want. In some ways, this is related to the 20 percent rule. It does not necessarily require one day a week (20 percent), but it does require some deliverable. The deliverable can be a new utility program that could be used by lots of different projects, it could be a new prototype of a new software product, or it could be an improvement for a business process that is used internally.

It also could imply providing the team member with a **high-end development environment**. For example, when building information visualization and virtual reality applications, special hardware and software environments can make it much easier to master the technology to develop the application.

Finally, today it is very important for team members to feel that what they are doing can make a difference. A team leader should encourage the team members to tackle problems that can make a difference in people's lives. This can easily be accomplished through the use of the 20 percent rule. Next figure lists some motivational don'ts to avoid demotivating team members.

Don'ts	Reasons
Assign unrealistic deadlines	Few people will work hard if they realize that a deadline is impossible to meet.
Ignore good efforts	People will work harder if they feel like their work is appreciated. Often, all it takes is public praise for a job well done.
Create a low-quality product	Few people can be proud of working on a project that is of low quality.
Give everyone on the project a raise	If everyone is given the same reward, then high-quality people will believe that mediocrity is rewarded—and they will resent it.
Make an important decision without the team's input	Buy-in is very important. If the project manager needs to make a decision that greatly affects the members of her team, she should involve them in the decision-making process.
Maintain poor working conditions	A project team needs a good working environment or motivation will go down the tubes. This includes lighting, desk space, technology, privacy from interruptions, and reference resources.

Source: Steve McConnell, *Adapted Rapid Development* (Redmond, WA: Microsoft Press, 1996).

Suggestions for motivation might include all of these EXCEPT:

- Setting realistic deadlines
- Giving all team members the same bonus on a project
- Recognize and reward good efforts
- Reward those with outstanding quality and effort
- Having a good working environment

Handling Conflict

The third component of staffing is organizing the project to minimize conflict among group members. *Group cohesiveness* (the attraction that members feel to the group and to other members) contributes more to productivity than do project members' individual capabilities or experiences.

Clearly defining the roles on the project and holding team members accountable for their tasks is a good way to begin mitigating potential conflict on a project. Some project managers develop a project charter, which lists the project's norms and ground rules. For example, the charter may describe when the project team should be at work, when staff meetings will be held, how the group will communicate with each other, and the procedures for updating the workplan as tasks are completed.

Conflict-Avoidance Strategies

- Clearly define plans for the project.
- Make sure the team understands how the project is important to the organization.
- Develop detailed operating procedures and communicate these to the team members.
- Develop a project charter.
- Develop schedule commitments ahead of time.
- Forecast other priorities and their possible impact on the project.

Source: H. J. Thamhain and D. L. Wilemon, "Conflict Management in Project Life Cycles," *Sloan Management Review* (Spring 1975).

ENVIRONMENT AND INFRASTRUCTURE MANAGEMENT

The environment and infrastructure management workflows support the development team throughout the development process. The environment workflow primarily deals with choosing the correct set of tools that will be used throughout the development process and identifying the appropriate set of standards to be followed during the development process.

Infrastructure management workflow deals with choosing the appropriate level and type of documentation that will be created during the development process.

CASE Tools

Computer-aided software engineering (CASE) is a category of software that automates all or part of the development process.

upper CASE

CASE software packages that are used primarily to support the analysis workflow to create integrated diagrams of the system and to store information regarding the system components

lower CASE

Support the design workflow that can be used to generate code for database tables and system functionality

Integrated CASE, or I-CASE

contains functionality found in both upper CASE and lower CASE tools in that it supports tasks that happen throughout the system-development process.

Benefits of using CASE

With CASE tools, tasks can be completed and altered much faster, development information is centralized, and information is illustrated through diagrams, which are typically easier to understand. Potentially, CASE can reduce maintenance costs, improve software quality, and enforce discipline. Some project teams even use CASE to assess the magnitude of changes to the project. Many modern CASE tools that support object-oriented systems development support a development technique known as round-trip engineering.

CASE repository

The central component of any CASE tool is the *CASE repository*, otherwise known as the information repository or data dictionary. The CASE repository stores the diagrams and other project information, such as screen and report designs, and it keeps track of how the diagrams fit together. For example, most CASE tools warn you if you place a field on a screen design that doesn't exist in your structural model. As the project evolves, project team members perform their tasks using CASE.

Standards

Members of a project team need to work together, and most project management software and CASE tools provide access privileges to everyone working on the system. When people work together, however, things can get pretty confusing. To make matters worse, people sometimes are reassigned in the middle of a project. It is important that their project knowledge does not leave with them and that their replacements can get up to speed quickly.

Types of Standards	Examples
Documentation standards	The date and project name should appear as a header on all documentation.
	All margins should be set to 1 inch.
	All deliverables should be added to the project binder and recorded in its table of contents.
Coding standards	All modules of code should include a header that lists the programmer, last date of update, and a short description of the purpose of the code.
	Indentation should be used to indicate loops, if-then-else statements, and case statements.
	On average, every program should include one line of comments for every five lines of code.
Procedural standards	Record actual task progress in the work plan every Monday morning by 10 AM.
	Report to project update meeting on Fridays at 3:30 PM.
	All changes to a requirements document must be approved by the project manager.
Specification requirement standards	Name of program to be created
	Description of the program's purpose
	Special calculations that need to be computed
	Business rules that must be incorporated into the program
	Pseudocode
	Due date
User interface design standards	Labels will appear in boldface text, left-justified, and followed by a colon.
	The tab order of the screen will move from top left to bottom right.
	Accelerator keys will be provided for all updatable fields.